Azure Retail Sales Data Engineering & Business Intelligence

PROJECT IMPLEMENTATION REPORT | November 2026

Azure Data Lake Storage Gen2 • Azure Synapse Analytics • Serverless SQL • Power BI

Portfolio project | Synthetic retail sales and customer datasets

Executive summary

This project implements a cloud-based retail analytics workflow that ingests synthetic sales and customer CSV data into Azure Data Lake Storage Gen2 (ADLS Gen2), orchestrates data movement with Azure Synapse pipelines, applies data-quality rules through serverless SQL views, enriches transactions with customer attributes, and delivers interactive Power BI reports. The completed workflow separates accepted sales from excluded records and supports regional, product and customer analysis.

Validated metric	Result
Raw sales rows	5,000
Customer rows	500
Accepted sales rows	4,920 (98.4%)
Excluded sales rows	80 (1.6%)
Rejected / quarantined breakdown	59 rejected / 21 quarantined
Clean sales revenue	£1,576,245.85
Power BI report pages	2

1. Project objectives

- Build a reproducible raw-to-staging-to-curated data pipeline using Azure services.
- Identify invalid, incomplete and anomalous sales records without silently deleting them.
- Enrich valid sales with customer segment and loyalty tier information.
- Provide interactive sales and customer dashboards with consistent financial totals.
- Demonstrate practical skills in cloud storage, orchestration, SQL, DAX, validation and BI reporting.

2. Technology stack

Layer	Technology	Purpose
Storage	Azure Data Lake Storage Gen2	Organised CSV files across raw, staging, processed and rejected zones
Orchestration	Azure Synapse pipeline (Copy Data)	Moves sales and customer files and materialises curated

		outputs
Transformation	Azure Synapse serverless SQL	External-file queries, data-quality classification, enrichment and reporting views
Analytics	Power BI Desktop / Power Query / DAX	Interactive dashboards, KPIs, region filters and monthly trends
Source control (planned)	GitHub	Versioned SQL, documentation and sanitised pipeline definitions

3. Solution architecture

Logical data flow:

> Synthetic sales & customers CSV → ADLS Gen2 /raw → Synapse Copy Data → ADLS Gen2 /staging
> → Synapse serverless SQL validation views → /processed (clean sales) + /rejected (excluded sales)
> → SQL enrichment & aggregation views → Power BI interactive reports

The serverless SQL views read data in storage and provide the transformation/query layer. The Copy Data activities write selected query results back to ADLS as CSV outputs. The customer staging branch is separate from the sales staging branch in the implemented pipeline.

4. Azure environment and storage organisation

Resource	Implemented value
Synapse workspace	synapseval29
Serverless SQL endpoint	synapseval29-ondemand.sql.azuresynapse.net
SQL database	RetailSalesDB
Storage account	valsynapsedataset1
Blob container	taxi-data
Project prefix	retail-project/
Storage zone	Example file / role
raw/	retail_sales_raw.csv; retail_customers.csv — source files
staging/	retail_sales_staging.csv; retail_customers_staging.csv
processed/	retail_sales_clean.csv — accepted records
rejected/	retail_sales_rejected.csv — excluded records

The storage container name reflects the existing lab environment; the retail-project prefix logically isolates the retail workload.

5. Source data and quality checks

The synthetic dataset contains 5,000 sales rows and 500 customer rows. The sales data supports transaction dates, product category, region, sales channel, payment method, quantity and GBP revenue. The customer file provides attributes used for segment and loyalty analysis.

The validation logic produced 4,920 accepted sales rows and 80 excluded rows. The excluded population was reported as 59 rejected and 21 quarantined records. Quality concerns investigated during development included invalid quantities, missing regions, invalid discounts and repeated transaction identifiers. A repeated transaction ID appeared across 21 records with differing transaction details; therefore, duplicate-ID handling requires business-rule awareness rather than indiscriminate removal.

Twenty otherwise accepted transactions had CustomerID C9999 with no matching customer row. They were retained in sales revenue and represented as Unknown customer segment/loyalty tier in reporting. This preserves the financial total while exposing the customer master-data gap.

6. Synapse pipeline implementation

Pipeline: PL_Retail_Sales_ETL

Activity	Source	Destination
Copy_Raw_Sales_To_Staging	Raw sales CSV	Staging sales CSV
Copy_Raw_Customers_To_Staging	Raw customers CSV	Staging customers CSV
Copy_Clean_Sales_To_Processed	dbo.vw_retail_sales_clean	Processed clean sales CSV
Copy_Rejected_Sales_To_Rejected	dbo.vw_retail_sales_rejected	Rejected sales CSV

The clean and rejected copy activities depend on successful sales staging. The customer staging copy runs independently. All four Copy Data activities were published and a fresh debug execution succeeded. File-to-file copy diagnostics report bytes transferred and file counts; these are not substitutes for row-count validation.

7. SQL transformation and reporting views

SQL view	Role
dbo.vw_retail_sales_staging	Read staged sales records
dbo.vw_retail_customers_staging	Read staged customer records
dbo.vw_retail_sales_clean	Select accepted sales
dbo.vw_retail_sales_rejected	Expose excluded sales
dbo.vw_retail_sales_enriched	Join accepted sales with customer attributes
dbo.vw_monthly_revenue	Aggregate revenue by year and month
dbo.vw_revenue_by_category	Aggregate revenue by product category
dbo.vw_revenue_by_region	Aggregate revenue by region
dbo.vw_revenue_by_customer_segment	Aggregate revenue by customer segment
dbo.vw_revenue_by_loyalty_tier	Aggregate revenue by loyalty tier

dbo.vw_revenue_by_payment_method	Aggregate revenue by payment method
dbo.vw_revenue_by_sales_channel	Aggregate revenue by channel

The analytical views were consumed in Power BI. To support interactive filtering across metrics and charts, the report additionally imported the transaction-level enriched view and built DAX measures on it.

8. Data validation and reconciliation

Validation	Observed result
Raw-to-output row reconciliation	5,000 = 4,920 clean + 80 excluded
Processed CSV line count	4,921 lines including header = 4,920 records
Customer staging count	500 rows
Unmatched accepted sales	20 rows, retained as Unknown
Revenue reconciliation	£1,576,245.85 across clean sales and summary views

Checks included COUNT(*), distinct customer IDs, comparisons across reporting views and region-filtered Power BI KPIs. Examples: North West — 647 transactions, 2,856 units, £212,628.74; South West — 608 transactions, 2,740 units, £204,047.34; Wales — 603 transactions, 2,694 units, £196,570.87.

9. Power BI semantic model and dashboard

Power BI imported seven reporting views initially, then added dbo.vw_retail_sales_enriched as an eighth query. Power Query created a MonthStartDate field for chronological monthly plotting and cleaned customer segment and loyalty labels, replacing null with Unknown. Interactive measures on the enriched table use the following DAX patterns:

```
Interactive Total Revenue = SUM('vw_retail_sales_enriched'[TotalAmountGBP])

Interactive Total Transactions = COUNTROWS('vw_retail_sales_enriched')

Interactive Total Units Sold = SUM('vw_retail_sales_enriched'[Quantity])
```

Sales Overview: total revenue, transactions, units sold, monthly revenue trend, revenue by product category, and revenue by region. Customer Insights: revenue by customer segment, loyalty tier, payment method and sales channel. Region slicers are synchronised between pages. The charts were switched from independent aggregated views to the enriched transaction-level table to enable consistent filtering.

Visual styling uses a light-grey page canvas, navy report headings, white chart/card backgrounds and consistent blue marks. The saved Power BI project is named Retail_Sales_Analytics_Dashboard.pbix.

10. Selected business insights

Dimension	Finding
Product category	Home was the highest-revenue category overall; Fitness was lowest
Customer segment	Consumer £574,150.50; Corporate £560,391.16;

	Small Business £436,539.84; Unknown £5,164.35
Loyalty tier	Gold £581,701.87; Silver £510,007.04; Bronze £479,372.59; Unknown £5,164.35
Sales channel	Online and In-store each contributed approximately £0.79M
Region	North West £212,628.74 and South West £204,047.34 were leading regions in the validated regional view

These findings describe a synthetic demonstration dataset and should not be presented as production retail performance.

11. Challenges and resolutions

- A leading space in an ADLS blob path prevented the intended SQL file reference from resolving; renaming the blob corrected the path.
- A linked service and datasets were lost after an unpublished session; they were recreated, published and revalidated.
- Power BI authentication to the Synapse serverless SQL endpoint initially failed; connection was eventually established and views loaded.
- A monthly chart initially combined months across years, then later repeated a regional total for each month; using a month-start date from the enriched transaction table resolved the issue.
- Independent aggregated views did not cross-filter with a Region slicer; rebuilding interactive visuals from the enriched table fixed the model behaviour.
- Null customer attributes appeared as (Blank); Power Query clean columns made the missing matches explicit as Unknown.

12. Limitations and future enhancements

- This is a synthetic, small-scale demonstration; operational scale and service-level performance have not been benchmarked.
- The pipeline uses Copy Data plus serverless SQL views; it is not a Spark-based distributed transformation pipeline.
- Add pipeline parameters, scheduling, retry policies, monitoring and alerting for production-like orchestration.
- Add automated validation assertions and a data-quality audit table, with explicit reject reason codes.
- Introduce date/customer/product dimension tables and a star schema for a more scalable BI semantic model.
- Version sanitised Synapse artifacts and SQL scripts in GitHub; add an architecture diagram and dashboard screenshots.
- Evaluate access controls, secret management and least-privilege deployment before any production use.

13. Portfolio deliverables and publication checklist

- Power BI report: Retail_Sales_Analytics_Dashboard.pbix (local saved artifact).
- SQL view scripts and data validation queries (export before GitHub publication).

- Sanitised Synapse pipeline JSON and dataset/linked-service templates (remove secrets and account-specific credentials).
- Screenshots: successful pipeline run, storage zones, representative SQL validation, Sales Overview and Customer Insights.
- README, architecture diagram and synthetic-data disclaimer.
- Do not publish storage keys, tokens, passwords, personal data or live connection credentials.

14. Conclusion

The project demonstrates a complete learning-oriented cloud analytics workflow from synthetic source data to curated outputs and interactive business reporting. It provides concrete evidence of ADLS Gen2 organisation, Synapse orchestration, serverless SQL transformation, data reconciliation, customer enrichment, Power Query preparation and DAX-driven dashboard interactivity. The next step is to version the implementation assets and screenshots in a documented GitHub repository.